

PROGRAM 9

ORGANIZING KUBERNETES PODS USING NAMESPACES AND LABELS

PROJECT STRUCTURE:

```
prog9/
├── namespaces.yaml
└── pods.yaml
```

STEP 1: Create namespaces.yaml file

STEP 2: Create pods.yaml file

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog9$ cat namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog9$ cat pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
```

STEP 3: Apply namespace and pod configurations

```
kubectl apply -f namespaces.yaml
```

```
kubectl apply -f pods.yaml
```

STEP 4: Verify namespaces

```
kubectl get namespaces
```

STEP 5: Verify pods in each namespace

```
kubectl get pods -n dev
```

```
kubectl get pods -n prod
```

STEP 6: Filter pods using labels

```
kubectl get pods -n dev -l app=frontend
```

```
kubectl get pods -n prod -l app=backend
```

STEP 7: Clean up resources

```
kubectl delete pods --all -n dev
```

```
kubectl delete pods --all -n prod
```

```
kubectl delete namespaces dev prod
```

OUTPUT:

Namespaces created successfully. Pods were organized using labels and verified using kubectl commands.

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog9$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, default-storageclass, dashboard

! /usr/bin/kubectl is version 1.28.15, which may have incompatibilities with Kubernetes 1.34.0.
  ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog9$ kubectl apply -f namespaces.yaml
kubectl apply -f pods.yaml
namespace/dev unchanged
namespace/prod unchanged
pod/frontend-pod unchanged
pod/backend-pod unchanged
pod/frontend-pod unchanged
pod/backend-pod unchanged
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog9$ kubectl get pods -n dev
kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   2 (29s ago) 41m
frontend-pod  0/1     Completed 1 (38m ago) 41m
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   2 (29s ago) 41m
frontend-pod  1/1     Running   2 (29s ago) 41m
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   2 (36s ago) 41m
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   2 (39s ago) 42m
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog9$ █
```

